

Assignment #3

Course: ISA 345

Instructor: Dr. Maliha Alam

Points:100

Submission instructions: this assignment is to be done individually. All your answers must be in a single document, *e.g.*, a *doc* or a *pdf* file. For database-modeling questions, you can take print screens of your diagrams and insert the resulting figures into your document. Another option is to export diagrams to figures by clicking at “File -> Data Modeler -> Print Diagram -> To Image File” inside Oracle SQL Developer, and then importing these figures into a text editor such as MS Word. Database-modeling questions must be answered using the Data Modeler tool inside Oracle SQL Developer (see Lecture #2). Reverse engineering is allowed (*i.e.*, creating diagrams from DDL/SQL code). All your SQL code must be well-formulated (*i.e.*, no errors) and sound (*i.e.*, it does what the question asks it to do). In particular, the grader must be able to copy your SQL code, paste it into an Oracle environment such as Oracle SQL Developer, and run the code without errors. Do not report images of your SQL code. Instead, report your code as text. You can copy your code from Oracle SQL Developer and paste it into a text editor such as MS Word. Code with errors or reported as images might receive zero points. Submit the final document on Canvas before the due date

Background story: After a successful internship at IMDB.com, you were hired by the Federal Aviation Administration (FAA) as a full-time database developer. Your primary job is to take care of a database containing data about all national flights departing from one of the airports in the NYC/Newark region (EWR, LGA, JFK). Since you are a new employee at FAA, it might be worthwhile to familiarize yourself with the relational database model shown on the following page. The underlying DDL (SQL code) is available on Canvas (file *CreateTables.ddl*). Recall that you can now generate database models from DDL files, and vice versa (see Lecture #10).

AIRLINE			
P *	Carrier	CHAR (2)	Two letter abbreviation
*	Name	VARCHAR2 (100)	Full name
AIRLINES_PK (Carrier)			

FLIGHT			
P *	FlightYear	SMALLINT	Date of departure
P *	FlightMonth	SMALLINT	Date of departure
P *	FlightDay	SMALLINT	Date of departure
	FlightActualDepTime	VARCHAR2 (5)	Actual departure time (local timezone)
P *	FlightScheduledDepTime	VARCHAR2 (5)	Scheduled departure time (local timezone)
	FlightDepDelay	SMALLINT	Departure delays in minutes. Negative times represent early departures
	FlightActualArrTime	VARCHAR2 (5)	Actual arrival time, local timezone
	FlightScheduledArrTime	VARCHAR2 (5)	Scheduled arrival time (local timezone)
	FlightArrDelay	SMALLINT	Arrival delays in minutes. Negative times represent early arrivals
PF *	FlightCarrier	CHAR (2)	Two letter carrier abbreviation
P *	FlightNumber	SMALLINT	Flight number
F *	TailNumber	VARCHAR2 (10)	Airplane's tail number
F	FlightOrigin	VARCHAR2 (5)	Airport origin
F	FlightDest	VARCHAR2 (5)	Airport destination
	FlightTime	SMALLINT	Amount of time spent in the air in minutes
	FlightDistance	SMALLINT	Distance flown
FLIGHT_PK (FlightNumber, FlightScheduledDepTime, FlightYear, FlightMonth, FlightDay, FlightCarrier)			
FLIGHT_AIRLINE_FK (FlightCarrier)			
FLIGHT_AIRPORT_FK (FlightOrigin)			
FLIGHT_AIRPORT_FKv1 (FlightDest)			
FLIGHT_PLANE_FK (TailNumber)			

PLANE			
P *	TailNum	VARCHAR2 (10)	Airplane's tail number
	PlaneYear	SMALLINT	Year manufactured
	PlaneType	VARCHAR2 (100)	Type of plane
	PlaneManufacturer	VARCHAR2 (100)	Plane's manufacturer
	PlaneModel	VARCHAR2 (100)	Plane's model
	PlaneNrEngines	SMALLINT	Number of engines
	PlaneNrSeats	SMALLINT	Number of seats
	PlaneEngine	VARCHAR2 (50)	Type of engine
PLANE_PK (TailNum)			

Flight's origin

Flight's destination

AIRPORT			
P *	FAACode	VARCHAR2 (5)	FAA airport code
*	AirportName	VARCHAR2 (100)	Usual name of the airport
*	AirportLatitude	NUMBER (9,6)	Geolocation of airport
*	AirportLongitude	NUMBER (9,6)	Geolocation of airport
*	AirportAltitude	SMALLINT	Altitude in feet
*	AirportTimezone	SMALLINT	Timezone offset from GMT
AIRPORT_PK (FAACode)			

Note on the two 1:M relationships between AIRPORT and FLIGHT:

Recall that every flight has an airport at the origin and another at the destination. The two relationships ensure that both airports are defined inside the table AIRPORT

Question 1: Your first task at FAA is to develop a series of queries that will help FAA and airports' stakeholders to improve their operations. You must report the SQL code (queries) that answers the questions below. Specifically, all questions must be answered using a single query, with the possible use of subqueries, without relying on further interpretation by a human. Although it is not strictly necessary, you might want to populate the tables AIRLINE, AIRPORT, PLANE, and FLIGHT. Populating these tables will help you to visualize the outcomes of your queries. Note, however, that although the background story is fictitious, this is a real-life dataset. That said, it might take you anywhere from some good minutes to a few hours to populate all the tables (there are more than 275,000 flights in the FLIGHT table). If you decide to populate the tables, please follow the steps below:

- 1) Drop all unused tables with the command **DROP TABLE <table_name> PURGE**; The option **PURGE** immediately gives you back the space used by the dropped tables.
- 2) Create the tables AIRLINE, AIRPORT, PLANE, and FLIGHT by dragging and dropping the file *CreateTables.ddl* into SQL Developer, and then running all the commands in it.
- 3) Run the INSERT commands inside the files *Airline_data.sql*, *Airport_data.sql*, *Planes_data.sql*, and *Flights_data.sql*. The INSERT commands in *Flights_data.sql* must be the last ones (why?).
- 4) It is highly advisable that you perform the above steps while on campus and connected to MU wireless or from a PC in the lab. By doing so, you will be directly connected to MU/FSB network, which means that all the commands will be executed faster. Using Remote Desktop Connection or VPN might cause a considerable delay.
- 5) Go get yourself some coffee/tea/coke/etc. and relax ... the whole process will take a while ☺

- a) FAA is trying to understand the factors that influence flight delays. One might argue that the time of the year is one of such factors. Focusing only on delayed flights at departure, write a query that counts the number of delays per month. **[7 points]**
- b) Everybody hates canceled flights! Write a query that lists the names of the airlines together with the number of times they canceled a flight. You can assume that a flight without an actual arrival time was canceled. Sort the results by the number of canceled flights in descending order. **[7 points]**
- c) Are new planes faster than old planes? Write a query that reports the average speed of planes by year of production. Remove missing data and sort the results by year (descending order). **[7 points]**
- d) You continue your efforts to understand the factors that influence flight delays. For example, you now hypothesize that flights to the West coast face higher delays than other flights. To test your hypothesis, write a query that reports airports' destination codes, destinations' time zone, and the number of delayed flights at departure. In particular, focus on flights that are delayed more than 1 hour and sort the results by the number of delays in descending order. **[9 points]**.
- e) FAA is considering penalizing airlines that have a very old fleet. Write a query that reports the average year of the planes owned by each airline. To help with visualization, sort the results by the average year in descending order. **[10 points]**
- f) In order to improve NY airports' infrastructure, FAA is interested in knowing the reason why some specific airplane models never operate in the NYC/Newark region. To help FAA, write a query that returns the model and manufacturers of the planes that have never departed from an airport in the NYC/Newark region (hint: recall that the whole table FLIGHTS is about flights departing from an airport in the NYC/Newark region). **[10 points]**

Question 2: Being an expert in database modeling, you suggest the following changes to the design of FAA's database:

- Co-ownership: one plane might (or might not) belong to multiple airlines (co-ownership), and an airline might (or might not) own multiple planes. **[15 points]**
- Weather forecasting: a weather forecasting company might sell forecasts (*i.e.*, wind speed, temperature, humidity, rain and snow precipitation per hour) to different airports, and one airport might purchase forecasts from multiple weather forecasting companies. Each data point about a weather forecasting company contains the company's name, address, and a unique identifier. **[15 points]**
- Social media platform: an airport might post information about flights using different social media platforms (see more info about posts below). Each social media platform might contain many posts from each airport. Each social media platform has an ID and a name. **[10 points]**
- Social media post: a single post on social media contains information about a single flight, but information about a single flight might be spread across several posts. Each social media post contains some text and a timestamp. **[10 points]**

Using Data Modeler, extend the initial data model on Page 2 by adding entities and relationships so as to capture the above business rules. Once again, recall that you can use the DDL file *CreateTables.ddl* available on Canvas to recreate the original model on Page 2 (see Lecture 10). Report the final model (*e.g.*, a screenshot).